cat = 1/3 0/3 = 0 3

T-8

D1 - cat chased mouse
D2 - dog barked cat
D3 - mouse ran cat
["barked", "cat", "chased", "dog", "mouse", "ran"]

	D1	D2	D3
barked	0	1/3 = 0.33	0
cat	1/3 = 0.33	1/3 = 0.33	1/3 = 0.33
chased	1/3 = 0.33	0	0
dog	0	1/3 = 0.33	0
mouse	0.33	0	0.33
ran	0	0	0.33

Normalized TT

	D1	D2	D3
barked	0	$1+\log_{10}(0.33)$	$0 = 0.086$
cat	$1+\log_{10}(0.33)$	$1+\log_{10}(0.33)$	$1+\log_{10}(0.33) = 0.051 = 0.66$
chased	$1+\log_{10}(0.33)$	0	$0 = 0.051 = 0$
dog	0	$1+\log_{10}(0.33)$	$0 = 0.051 = 0$
mouse	$1+\log_{10}(0.33)$	0	$1+\log_{10}(0.33) = 0.051 = 0$
ran	0	0	$1+\log_{10}(0.33) = 0.051 = 0.086$

IDF

barked $\log(\frac{3}{1+1}) = \log(\frac{3}{2}) = 0.17$
cat $\log(\frac{3}{1+3}) = \log(\frac{3}{4}) = -0.12$
chased $\log(\frac{3}{1+1}) = \log(\frac{3}{2}) = 0.17$
dog $\log(\frac{3}{1+1}) = \log(\frac{3}{2}) = 0.17$
mouse $\log(\frac{3}{1+2}) = \log(\frac{3}{2}) = 0.17$
ran $\log(\frac{3}{1+1}) = \log(\frac{3}{2}) = 0.17$

TF x IDF

	D1	D2	D3
barked	0 x 0.17	0.51 x 0.17	0 x 0.17
cat	0.51 x (0.12) = 0.064	-0.064	-0.064
chased	0.086	0	0
dog	0	0.086	0
mouse	0	0	0
ran	0	0	0.086

D1 [cat, chased, mouse]
D2 [dog, barked, cat]
D3 [mouse, ran, cat]

D1

context(words)	context(index)	target word	target index
chased	2	Cat	1
cat, mouse	1, 4	chased	2
chased	2	mouse	4

D2

context word	content(index)	target word	target
barked	0	dog	3
dog, cat	3, (1, 1)	barked	0
barked	0	cat	1

D3

context word	Context index	target word	Target ind
ran	5	mouse	4
mouse, cat	4, 1	ran	5
ran	5	cat	1

1) What is the intuition behind using TF-IDF instead of raw term frequency?

- Balancing importance, it balances how often a word appears in a single document against how common it is across the entire corpus
- Filtering noise, raw frequency (TF) overvalues common words like "the, is" that carry little meaning. TF-IDF penalizes these words to highlight more descriptive te

2) Which terms receive higher importance in TF-IDF representations?

- Terms that appear frequently in a single document but rarely across the entire corpus receive the highest importance

3) How would the representation change if you added more documents to the corpus?

- Impact of adding more documents,

Recalculated IDF - As total number of document increase th IDF values for every word in the vocabulary change

Vocabulary growth - New documents may introduce new unique words increasing the size and sparsity of the final TF-IDF matrix.

4) How does CBOW handles this?
- CBOW handles this by averaging projection of all context words into single hidden layer vector, this allows model to compress multiple contexts into a dense representation that captures the word's meaning.

5) Can you connect idea to Distributional Hypothesis?
- Yes, CBOW directly implements distributional hypothesis by learning word meanings from their surrounding context.